

Linear Regression Basics

MDragon Data Tools - Study Guide

What is Linear Regression?

Linear regression models the relationship between:

- **Dependent variable (Y):** What we're trying to predict
- **Independent variable (X):** What we use to make predictions

The Regression Line

$$\hat{y} = b_0 + b_1x$$

where:

$\hat{y}$ = predicted value

b_0 = y-intercept

b_1 = slope

x = independent variable

Key Formulas

Slope (b_1)

$$b_1 = r \times (s^y / s^x)$$

or

$$b_1 = \Sigma[(x-\bar{x})(y-\bar{y})] / \Sigma(x-\bar{x})^2$$

Intercept (b_0)

$$b_0 = \bar{y} - b_1\bar{x}$$

Correlation Coefficient (r)

$$r = \frac{\Sigma[(x-\bar{x})(y-\bar{y})]}{\sqrt{[\Sigma(x-\bar{x})^2 \times \Sigma(y-\bar{y})^2]}}$$

Range: $-1 \leq r \leq 1$

Coefficient of Determination (R^2)

$$R^2 = r^2$$

Interpretation: % of variance in Y explained by X

Example: Step-by-Step

 **Problem:** Study hours (X) vs exam scores (Y):

Hours	Score
2	65
4	75
5	82
6	85
8	92

💡 **Solution:**

1. Calculate means:

$$\bar{x} = (2+4+5+6+8)/5 = 5$$
$$\bar{y} = (65+75+82+85+92)/5 = 79.8$$

2. Calculate slope:

$$\Sigma(x-\bar{x})(y-\bar{y}) = (-3)(-14.8) + (-1)(-4.8) + (0)(2.2) + (1)(5.2) + (3)(12.2) = 91$$
$$\Sigma(x-\bar{x})^2 = 9 + 1 + 0 + 1 + 9 = 20$$
$$b_1 = 91/20 = 4.55$$

3. Calculate intercept:

$$b_0 = 79.8 - 4.55(5) = 57.05$$

4. Regression equation:

$$\hat{y} = 57.05 + 4.55x$$

Interpretation:

- For each additional hour studied, expect score to increase by 4.55 points
- Starting point (0 hours): expect 57.05

Predictions

Using $\hat{y} = 57.05 + 4.55x$:

Predict score for 7 hours:

$$\hat{y} = 57.05 + 4.55(7) = 88.9$$

Assumptions

1. Linearity (relationship is linear)
 2. Independence of errors
 3. Homoscedasticity (constant variance)
 4. Normality of residuals
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Use our calculators to perform regression analysis instantly!

MDragon Data Tools | Statistics Education Platform

mondragon-developer.github.io/statools